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Assignment 2

Linux Security Basics

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1.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo apt update
33 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

   This command retrieves all the available packages and updates in Ubuntu.

2.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo apt upgrade -y
Restarting services...
 systemctl restart cups.service geoclue.service multipathd.service open-vm-tools
.service packagekit.service vgauth.service
Service restarts being deferred:
 systemctl restart NetworkManager.service
 systemctl restart gdm3.service
 systemctl restart unattended-upgrades.service
 systemctl restart user@1000.service

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

   This command installs the newest versions of all packages on my Ubuntu server.

3.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo reboot
```

This restarts the system to allow updates to take effect.

4.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo su root
```

The prompt becomes:

```
root@ubuntuserver-lab:/home/kadetj/Desktop#
```

This command opens the root.

5.
```
root@ubuntuserver-lab:/home/kadetj/Desktop# useradd bobby
root@ubuntuserver-lab:/home/kadetj/Desktop# adduser sally
Adding user `sally' ...
Adding new group `sally' (1002) ...
Adding new user `sally' (1002) with group `sally' ...
Creating home directory `/home/sally' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sally
Enter the new value, or press ENTER for the default
        Full Name []: Sally
        Room Number []: 1
        Work Phone []: 845456789
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
```

The useradd command is very basic, and a base level add while the adduser command goes much more in depth by adding a home directory, a password, and other information.

6.
```
root@ubuntuserver-lab:/home/kadetj/Desktop# su - sally
sally@ubuntuserver-lab:~$ S
```

This command switches to and loads into sally's environment.

7.
```
sally@ubuntuserver-lab:~$ adduser earl
adduser: Only root may add a user or group to the system.
```

It does not allow the user to be created because sally does not have permissions to do this function. You must add sally to the group that's allowed to do this. To do this you would use this command: sudo usermod -aG sudo sally.

8.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo userdel bobby
```

This command deletes an added user.

9.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo passwd sally
passwd: password updated successfully
```

This command allows you to change the password of a specific user.

10. It is bad practice to stay logged in to the root because the root has full access so if there are attackers, they can easily access everything through the root shell and the system will be compromised.

11.
```
kadetj@ubuntuserver-lab:~/Desktop$ id
uid=1000(kadetj) gid=1000(kadetj) groups=1000(kadetj),4(adm),24(cdrom),27(sudo),
30(dip),46(plugdev),110(lxd)
```
This command shows you your uid, gid, and groups.

12.
```
kadetj@ubuntuserver-lab:~/Desktop$ groups ubuntu
```
My ubuntu is not a user in the system, so it does not belong to any groups.

13.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo usermod -aG sudo sally
kadetj@ubuntuserver-lab:~/Desktop$ su - sally
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

sally@ubuntuserver-lab:~$ sudo adduser earl
[sudo] password for sally:
Adding user `earl' ...
Adding new group `earl' (1001) ...
Adding new user `earl' (1001) with group `earl' ...
Creating home directory `/home/earl' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for earl
Enter the new value, or press ENTER for the default
        Full Name []: Earl
        Room Number []: 2
        Work Phone []: 8456789033
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
```
The command sudo usermod -aG sudo sally gives sally permissions by adding her to the sudo group, then we switch into sally's environment by using su – sally, and finally we add earl using adduser.

14.
```
sally@ubuntuserver-lab:~$ exit
logout
kadetj@ubuntuserver-lab:~/Desktop$ sudo groupadd cybersec
kadetj@ubuntuserver-lab:~/Desktop$
```
This command adds a new group named cybersec.

15.
```
kadetj@ubuntuserver-lab:~/Desktop$ sudo usermod -aG cybersec sally
```
This command adds sally to the group cybersec.

16.
```
kadetj@ubuntuserver-lab:~/Desktop$ groups sally
sally : sally sudo cybersec
```

```
kadetj@ubuntuserver-lab:~/Desktop$ id sally
uid=1002(sally) gid=1002(sally) groups=1002(sally),27(sudo),1003(cybersec)
```

Both commands “groups” and “id” can show you which groups sally belongs to, but for just searching groups the command “groups” is more straight forward due to that being the only information outputted, on the other hand the command “id” shows the groups plus the added information of sally’s uid and gid.

17.
```
kadetj@ubuntuserver-lab:~/Desktop$ mkdir lab1
kadetj@ubuntuserver-lab:~/Desktop$ ls -ld lab1
drwxrwxr-x 2 kadetj kadetj 4096 Mar  1 02:12 lab1
```

The command mkdir lab1 makes the directory lab1 and then the command ls –ld lab1 shows the permissions of the directory. The owner is kadetj and the group owner is kadetj. The owner and group owner both have read, write, and execute permissions. Other has read and execute permissions.

18.
```
kadetj@ubuntuserver-lab:~/Desktop$ cd lab1
kadetj@ubuntuserver-lab:~/Desktop/lab1$ nano helloWorld
kadetj@ubuntuserver-lab:~/Desktop/lab1$ chmod +x helloWorld
kadetj@ubuntuserver-lab:~/Desktop/lab1$ ./helloWorld
Hello World!
```

The command cd lab1 changes the directory to lab1, then nano helloWorld creates a file named helloWorld, next we edit the file and save it, after that we give the file executable permissions, and lastly, we run the file.

19.
```
kadetj@ubuntuserver-lab:~/Desktop/lab1$ chmod g+wx helloWorld
kadetj@ubuntuserver-lab:~/Desktop/lab1$ ls -l helloWorld
-rwxrwxr-x 1 kadetj kadetj 32 Mar  1 02:21 helloWorld
```

The command chmod g+wx helloWorld gives the group owner of the helloWorld file write and execute permissions. Then I use ls –l helloWorld to check the permissions of the file. The owner and group owner both have read, write, and execute permissions. Other has only read and execute permissions.

20.
```
kadetj@ubuntuserver-lab:~/Desktop/lab1$ getfacl helloWorld
# file: helloWorld
# owner: kadetj
# group: kadetj
user::rwx
group::rwx
other::r-x
```

This command shows the access control list entries. It lists the file name, the owner name, the group owner name, the user permissions, the group owner permissions, and the other permissions.

21.
```
kadetj@ubuntuserver-lab:~/Desktop/lab1$ setfacl -m u:sally:rw helloWorld
```

This command grants sally permission to read and write in the helloWorld file.